# PING PONG IN $\mathrm{PGL}_2(\mathbb{Z}[\frac{1}{2}])$

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Abstract. A two-page example for alphaTeX: the ping-pong lemma, Sanov's free subgroup of $\mathrm{SL}_2(\mathbb{Z})$, and why $\mathrm{PGL}_2(\mathbb{Z}[\frac{1}{2}])$ is discrete only in a product of two completions.

## 1. Introduction

Given finitely many matrices $S \subseteq \mathrm{PGL}_2(\overline{\mathbb{Q}})$, what is the group $\langle S \rangle$ they generate? The first tool for showing that a group is free is the ping-pong lemma (Lemma 2.1). Tits used it to show that a finitely generated linear group either contains a free subgroup of rank two or is virtually solvable [3].

We give the classical example of Sanov [1] in Section 2, and in Section 3 we look at a group that is not discrete at any single place.

## 2. Ping pong

**Lemma 2.1** (Ping pong)**.** *Let $a$ and $b$ act on a set $X$, and let $X_a, X_b \subseteq X$ be disjoint and nonempty. Suppose $a^n(X_b) \subseteq X_a$ and $b^n(X_a) \subseteq X_b$ for every $n \neq 0$. Then $\langle a, b \rangle$ is free on $a$ and $b$.*

*Proof.* A nontrivial reduced word in $a$ and $b$ is conjugate to one of the form $w = a^{n_k} b^{m_{k-1}} \cdots b^{m_1} a^{n_1}$ with every exponent nonzero. Each letter moves $X_b$ into $X_a$ or $X_a$ into $X_b$ in turn, so $w(X_b) \subseteq X_a$. Since $X_a \cap X_b = \emptyset$, $w$ is not the identity. □

**Theorem 2.2** (Sanov)**.** *The matrices*

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$

*generate a free subgroup of $\mathrm{SL}_2(\mathbb{Z})$ of rank two.*

*Proof.* Let $\mathrm{SL}_2(\mathbb{Z})$ act on $\mathbb{R} \cup \{\infty\}$ by Möbius transformations, and set $X_A = \{x : |x| > 1\}$ and $X_B = \{x : |x| < 1\}$. If $|x| < 1$ then $|A^n x| = |x + 2n| > 2|n| - 1 \geq 1$, so $A^n(X_B) \subseteq X_A$. If $|x| > 1$ then $B^n x = x/(2nx + 1)$ and $|1/B^n x| = |2n + 1/x| > 1$, so $B^n(X_A) \subseteq X_B$. Apply Lemma 2.1. □

*Date*: Demo document for the AlphaCentaur talk.

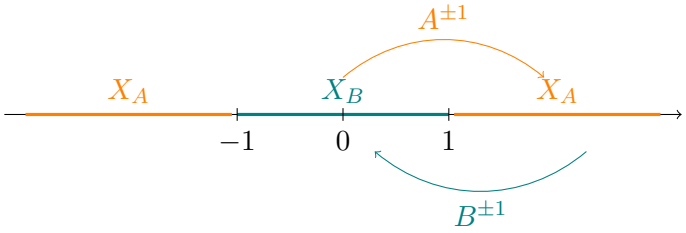



FIGURE 1. The ping-pong sets for Theorem 2.2.

## 3. A group that needs two places

**Proposition 3.1.** *The group* $\Gamma = \mathrm{PGL}_2(\mathbb{Z}[\frac{1}{2}])$ *is not discrete in* $\mathrm{PGL}_2(\mathbb{R})$ *or in* $\mathrm{PGL}_2(\mathbb{Q}_2)$*, but it is discrete in* $\mathrm{PGL}_2(\mathbb{R}) \times \mathrm{PGL}_2(\mathbb{Q}_2)$*.*

*Proof.* The ring $\mathbb{Z}[\frac{1}{2}]$ is dense in $\mathbb{R}$, so the upper triangular unipotent matrices in $\Gamma$ are not discrete in $\mathrm{PGL}_2(\mathbb{R})$. In $\mathrm{PGL}_2(\mathbb{Q}_2)$, the infinite group $\mathrm{PGL}_2(\mathbb{Z})$ lies in the compact group $\mathrm{PGL}_2(\mathbb{Z}_2)$, so $\Gamma$ is not discrete there either. Finally, $\mathbb{Z}[\frac{1}{2}]$ is discrete in $\mathbb{R} \times \mathbb{Q}_2$, and so $\Gamma$ is discrete in the product; see [2, Chapter II], where $\Gamma$ acts on the product of the hyperbolic plane and the 3-regular tree. □

*Remark* 3.2. Proposition 3.1 is why Sanov's group of Theorem 2.2 is *geometric* while $\mathrm{PGL}_2(\mathbb{Z}[\frac{1}{2}])$ is *arithmetic* in the sense of the Classification Conjecture.

## References

[1] I. N. Sanov, *A property of a representation of a free group*, Doklady Akad. Nauk SSSR (N.S.) **57** (1947), 657–659.
[2] J.-P. Serre, *Trees*, Springer-Verlag, Berlin, 1980.
[3] J. Tits, *Free subgroups in linear groups*, J. Algebra **20** (1972), 250–270.